

AI Financial Crime Investigation Copilot (AML)

Q1:

- * What triggers the alert?
- * What is the end-to-end investigation flow?
- * What red flags would the investigator look for?
- * What data/fields would be needed?
- * What investigation questions should AI answer?
- * What evidence should be documented?
- * What recommended next steps should AI provide?

Response:

I recommend proposing “**Unusual Incoming Funds Followed by Rapid Transfers to Multiple Beneficiaries.**”

The investigation would assess whether a personal account is acting as a **possible money mule or pass-through account**. This is a well-grounded scenario: the UK Financial Conduct Authority identifies unexplained deposits, rapid dispersal across accounts, and activity inconsistent with a customer’s profile as indicators worth investigating. An alert alone does not establish money laundering. [FCA](#)

The following is a **proposed prototype workflow**, using fictional Indian banking data. Its numeric thresholds are design assumptions for discussion, not regulatory requirements.

Example scenario

A salaried customer usually receives ₹80,000 per month and makes routine household payments. Over one day, the account receives **₹12 lakh from four unfamiliar senders**. Within six hours, **₹10.8 lakh is transferred to six beneficiaries**, five of whom were added recently.

The investigation question is: **Does this activity have a credible, documented purpose, or does it warrant further financial-crime investigation?**

1. What triggers the alert?

For the prototype, an illustrative rule could require all three conditions:

- Incoming funds over 24 hours exceed five times the customer’s average monthly credits over the preceding six months.
- Outgoing transfers within 24 hours equal at least 80% of those incoming funds.
- Transfers go to at least three distinct beneficiaries.

Recently added beneficiaries would strengthen the alert's priority.

In the example, incoming funds are **15 times** the historical monthly average, and outgoing transfers equal **90%** of incoming funds.

These calculations describe account activity. They do not prove that particular incoming funds financed particular outgoing payments, especially where the account already held a balance.

2. What is the end-to-end investigation flow?

This follows the P1-P11 structure already established in your capstone.

Step	Investigation activity	Expected output
P1	Receive the alert and validate triggering transactions	Alert record with transaction references
P2	Check completeness, priority, and related alerts	Investigation assignment and missing-data list
P3	Review customer profile, KYC, expected activity, and previous cases	Customer context and historical baseline
P4	Reconstruct incoming and outgoing payments chronologically	Timeline, activity ratios, and balance movements
P5	Examine senders, beneficiaries, and available account connections	Relationship map with supported links
P6	Review existing screening results and approved adverse-media findings	Relevant findings and unresolved matches
P7	Retrieve applicable internal investigation procedures	Policy references and required checks
P8	Assemble evidence and draft the case narrative	Summary, open questions, and proposed next steps
P9	Validate calculations, citations, and completeness	Quality findings and corrections
P10	Investigator assesses explanations and decides the disposition	Further investigation, supported closure, or escalation
P11	Record the decision, rationale, and approvals	Complete case record and audit trail

3. What red flags would the investigator look for?

The principal sourced indicators are unexplained receipts, rapid onward movement, and transactions inconsistent with expected customer activity. [FCA](#)

For this scenario, the proposed checks are:

- Multiple senders with no established relationship to the customer.
- Most funds leaving soon after receipt.
- Several recently added beneficiaries.
- Repeated similar patterns on earlier days.

- Little apparent economic purpose for receiving and forwarding the money.
- Inconsistent explanations or documents.
- Relevant previous alerts or verified fraud complaints, where available.

Investigators should also test legitimate explanations, such as a property transaction, family settlement, or documented repayment. Large value and speed alone are insufficient grounds for a conclusion.

4. What data and fields would be needed?

Data category	Minimum fields
Alert	Alert ID, scenario, trigger time, rule version, triggering transaction IDs
Customer/KYC	Customer ID, occupation, declared income, expected activity, risk rating, KYC review date
Account	Account ID, customer ID, opening date, account type, balance history
Transactions	Transaction ID, timestamp, direction, amount, currency, status, channel, sender/beneficiary reference, payment description
Beneficiaries	Beneficiary ID, bank/account reference, addition date, previous payment history
Historical activity	Six months of transactions, monthly credit totals, usual senders and beneficiaries
Prior cases	Alert/case ID, relevant findings, disposition, decision date
Supporting evidence	Customer explanation, documents, screening results, applicable policy versions

For the first prototype, use **synthetic data for one customer and one account**. Include identifiers for external counterparties, but only show onward transfers where the dataset actually provides them. AI should flag unavailable information rather than invent it.

5. What investigation questions should AI answer?

- What changed compared with the customer's normal activity?
- Who sent the money, and are these established senders?
- How much left the account, how quickly, and to whom?
- Which beneficiaries are new?
- Has this pattern occurred before?
- What relationships between accounts are supported by available evidence?
- Does the stated purpose explain both the receipts and onward transfers?
- What evidence supports a legitimate explanation?
- What information is missing or contradictory?
- Which policy checks remain incomplete?

Every answer should identify its supporting records and distinguish **observed facts, possible explanations, and unresolved questions**.

6. What evidence should be documented?

The proposed case package should contain:

- The alert rule, version, and exact triggering transactions.
- Historical baseline and calculation method.
- Transaction timeline, opening balance, receipts, payments, and closing balance.
- Beneficiary addition dates and relevant payment history.
- A relationship map with evidence for each connection.
- Customer explanations and supporting documents, where obtained through an authorized process.
- Evidence supporting both suspicious and legitimate interpretations.
- Data gaps, screening findings, policy references, and investigator rationale.

The FCA also emphasizes documenting investigation outcomes and reasoning, supporting the need for an evidence-based case record. [FCA](#)

7. What recommended next steps should AI provide?

Finding	Proposed AI recommendation
Important records are missing	Obtain the specified records before concluding
A credible explanation needs corroboration	Request supporting documents through the approved investigator workflow
Available evidence supports a legitimate purpose	Recommend closure for human review, with rationale
Significant indicators remain unexplained	Recommend escalation to the designated AML/compliance team
Evidence suggests fraud or account compromise	Recommend coordination with the fraud team
Potential reporting considerations arise	Refer to authorized compliance staff under applicable policy

The prototype should produce recommendations for review. Account restrictions, case disposition, and regulatory reporting decisions remain with authorized personnel.

For the meeting, I would propose **one suspicious case and one legitimate look-alike case** using the same alert pattern. That lets the team assess whether the copilot can explain why an alert fired, weigh competing explanations, identify missing evidence, and recommend a defensible next step—without coding yet.

Q2: Can you provide test data and steps for one fraud case and one legitimate look alike case

Response:

Below are **two fictional test cases with the same transaction pattern**: one fraud-linked case and one legitimate look-alike. Use them to check whether the copilot evaluates supporting evidence rather than treating an alert as proof of wrongdoing.

All names, IDs, amounts, and documents are synthetic. The “actual outcome” is a test label for your team; **do not provide that label to the AI during investigation**.

1. Common transaction pattern

Both customers have this historical profile:

Field	Test value
Account type	Personal savings
Account age	Four years
Occupation	Salaried employee
Declared monthly salary	₹80,000
Average monthly credits over previous six months	₹80,000
Opening balance on investigation day	₹20,000
Previous unusual activity	None
KYC status	Current
Sanctions/PEP screening	No confirmed matches

Use the following transactions for **both cases**, substituting the appropriate account and counterparty IDs.

Transaction ID	Time	Direction	Counterparty	Amount	Balance after transaction
T001	09:00	Credit	S01	₹3,00,000	₹3,20,000
T002	09:20	Credit	S02	₹3,00,000	₹6,20,000
T003	09:40	Credit	S03	₹3,00,000	₹9,20,000
T004	10:00	Credit	S04	₹3,00,000	₹12,20,000
T005	10:15	Debit	B01	₹1,80,000	₹10,40,000
T006	10:30	Debit	B02	₹1,80,000	₹8,60,000
T007	10:45	Debit	B03	₹1,80,000	₹6,80,000

Transaction ID	Time	Direction	Counterparty	Amount	Balance after transaction
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T008	11:00	Debit	B04	₹1,80,000	₹5,00,000
T009	11:15	Debit	B05	₹1,80,000	₹3,20,000
T010	11:30	Debit	B06	₹1,80,000	₹1,40,000

Assume all transactions are successful, in INR, and processed through bank transfers. Beneficiaries B01–B05 were added the previous day; B06 was added six months earlier.

Expected alert calculations:

- Incoming funds: **₹12,00,000**
- Historical monthly average: **₹80,000**
- Incoming activity multiplier: **15×**
- Outgoing funds: **₹10,80,000**
- Outgoing-to-incoming ratio: **90%**
- Distinct outgoing beneficiaries: **6**
- Time from first receipt to last outgoing transfer: **2 hours 30 minutes**

These satisfy the illustrative prototype alert rule from our earlier proposal.

2. Case A — Fraud-linked pass-through account

Test label: Funds received from investment-scam victims and forwarded to accounts controlled by the scam network.

Field	Synthetic test data
Case ID	CASE-F001
Customer ID	CUST-F001
Account ID	ACC-F001
Customer name	Arjun Mehta
Sender relationship	Four unrelated individuals
Payment descriptions	“Investment allocation”
Beneficiary relationship	Customer cannot explain the relationships

Customer’s initial explanation “I received money from friends and repaid personal loans.”

Provide these evidence records in stages:

Evidence ID	Evidence supplied	Investigation significance
F-E01	Sender S01’s bank confirms a verified investment-scam	Connects one receipt to

Evidence ID	Evidence supplied	Investigation significance
	complaint linked to T001	reported fraud
F-E02	S02 provides messages showing instructions to send ₹3 lakh to ACC-F001 for an investment	Corroborates the alleged source and purpose
F-E03	Customer initially describes all senders as friends, then admits not knowing S01 or S02	Contradicts the initial explanation
F-E04	No loan agreements or prior loan receipts support the claimed repayments	Leaves outgoing payments unexplained
F-E05	An approved internal record links B01 to an earlier confirmed fraud case	Adds a documented counterparty concern
F-E06	Customer provides messages instructing them to forward funds and retain ₹1.2 lakh	Supports possible facilitation of the scheme

Investigation steps and expected results

Step	Action	Expected copilot result
1	Load profile and transactions only	Explain the alert and identify questions; avoid concluding fraud
2	Request sender relationships and payment purpose	Identify the mismatch between investment descriptions and claimed personal transfers
3	Add F-E01 and F-E02	Identify the supported fraud connection, citing the relevant receipts
4	Add customer responses and F-E04	Document inconsistencies and missing corroboration
5	Add F-E05 and F-E06	Explain the combined evidence and remaining uncertainty
6	Generate investigation package	Recommend escalation to AML/compliance and coordination with the fraud team

Expected final recommendation: Escalate for authorized investigation. Document the evidence linking receipts to reported fraud, onward transfers, and inconsistent explanations. Assess whether the customer knowingly participated, was deceived, or experienced account compromise.

The AI should not independently declare criminal guilt, freeze the account, or submit a regulatory report.

3. Case B — Legitimate family settlement

Test label: A documented family settlement explains the same receipts and rapid onward transfers.

Field	Synthetic test data
Case ID	CASE-L001

Field	Synthetic test data
Customer ID	CUST-L001
Account ID	ACC-L001
Customer name	Neeraj Shah
Sender relationship	Four relatives contributing to a family settlement
Payment descriptions	"Family settlement contribution"
Beneficiary relationship	Six relatives entitled to settlement payments
Customer's explanation	"I coordinated payments under a signed family settlement agreement."
Provide these supporting records:	

Evidence ID	Evidence supplied	Investigation significance
L-E01	Signed settlement agreement dated before the transfers	Establishes an independently reviewable purpose
L-E02	Agreement lists S01–S04 contributing ₹3 lakh each	Reconciles all incoming payments
L-E03	Agreement lists B01–B06 receiving ₹1.8 lakh each	Reconciles all outgoing payments
L-E04	Agreement allocates the remaining ₹1.2 lakh to the customer	Explains the retained amount
L-E05	Approved verification confirms participant identities and account details match the agreement	Corroborates counterparties
L-E06	Supporting sender records show the contributions came from established personal savings	Supports the stated source of funds
L-E07	Customer explanation remains consistent; no relevant fraud complaint or prior adverse case is found in available records	Provides additional context

Investigation steps and expected results

Step	Action	Expected copilot result
1	Load profile and transactions only	Raise the same alert as Case A
2	Request purpose and counterparty relationships	Identify the settlement explanation as a hypothesis requiring evidence
3	Add L-E01–L-E04	Reconcile ₹12 lakh received, ₹10.8 lakh distributed, and ₹1.2 lakh retained

Step	Action	Expected copilot result
4	Add L-E05 and L-E06	Check identity, account, date, and source-of-funds consistency
5	Review L-E07 and outstanding gaps	Explain why the initial indicators have a supported legitimate explanation
6	Generate investigation package	Recommend closure for investigator approval, with evidence references

Expected final recommendation: Close the alert following human verification of the settlement and supporting records. Record why the unusual activity is reasonably explained.

Absence of sanctions matches or fraud complaints alone should not justify closure.

4. How to compare the AI's performance

Check	Expected result
Alert detection	Both cases trigger the same rule
Calculation accuracy	Both produce 15× incoming activity and a 90% outgoing ratio
Evidence handling	Every material conclusion cites a supplied record
Initial judgment	Neither case is classified as confirmed fraud from transactions alone
Fraud case recommendation	Escalation based on corroborating evidence
Legitimate case recommendation	Human-reviewed closure based on verified reconciliation
Missing information	AI explicitly identifies unavailable evidence
Human control	AI recommends; the investigator decides

Useful negative test: Remove L-E05 and L-E06 from the legitimate case. The copilot should request verification rather than confidently recommend closure solely because a settlement document exists.